



ZHANG JIANNAN

M.Sc.

@ jiannan.zhang@tum.de

+49-176-630-81506

Munich, Germany

jiannan-zhang

STRENGTHS

Modelling and Simulation

Flight Dynamics and Control

Control Theory

Control Allocation Optimization

DO-331 Model Based Design

Safety Critical Development

Object Oriented Programming

Nonlinear System Trim Linearization

Flight Testing and Operation

UAV Piloting

SOFTWARE SKILLS

MATLAB/Simulink

Simulink Model Advisor

Control Design Toolbox

Math Symbolic Toolbox

Git

GitLab

Polarion

C/C++

Eclipse

CMake

Visual Studio Code

Python

Solidworks

LANGUAGES

English: Professional Working

Chinese: Native

German: Goethe-Zertif. B1

OBJECTIVE

With years of experience in the field of flight dynamics modelling, control and simulation in both industry and academy, I am motivated to look for challenging opportunities where both my mindset for innovation and sound skills for safety critical system development can contribute, to exploit the promising future of aerospace and air mobility industry.

PROFESSIONAL EXPERIENCE

Research Associate | Institute of Flight System Dynamics

06.2016 – Now

Munich, Germany

- Participate in flight control system development for UAV prototypes.
- Lead of control allocation development for both general and safety critical systems.
- Organize and participate flight tests for small UAVs and work as test pilots.
- Support on building system modelling and analysis tool chain.

Intern Software Development | Phoenix-Wings GmbH

06.2020 – Now

Ismaning, Germany

- Focus on building a linearization tool in C++ to linearize an eVTOL dynamic model.
- Support in testing the software tool chains developed in the company including model-in-the-loop and software-in-the-loop simulation and ground control system.
- Attend flight tests and support in-flight ground control software testing.
- Support in preparing documents and communication with Chinese customers.

Flight Control Engineer | Autel Robotics Europe GmbH

06.2016 – 01.2020

Ismaning, Germany

- Participate in development of flight control system for a 5kg tilt-rotor eVTOL UAV.
- Focus on the attitude loop design and control allocation for the tilting mechanism.
- Organize flight tests and pilot the aircraft with aid of autopilots.
- Responsible for organizing administrative communication and reports to the Chinese headquarter of the project.

EDUCATION

Doctoral Candidate | Technical University of Munich (TUM)

06.2016 – Now (Expected Grad.: 2022)

Munich, Germany

- Research Direction:** Redundant System Control Allocation Optimization & Protection

M.Sc. Aerospace Engineering | Nanyang Technological University (NTU)

08.2013 – 02.2016

Singapore

- CGPA:** 4.92 (CAP: 1.06) **Major Subjects:** Flight Dynamics, Light Weight Structures
- ### B.Eng. Aircraft Design Engineering | Beijing Institute of Technology (BIT)

08.2008 – 07.2012

Beijing, China

PROJECTS

V280 - 280kg eVTOL with Hybrid Configuration | [TUM-FSD](#)

📅 02.2020 – Now

DEV Industry Project

- **Role:** Lead of control allocation development
- **Main Contribution:** Develop the control allocation algorithm for the complete flight envelope (hover, transition and cruise flight). In addition to nominal operation, the algorithm can safely operate with a critical hover propeller failure given information from fault detection and monitoring scheme.
- **Development:** Strictly under code-compliant guidelines specified by DO-331 MBD design process (standard libraries, Model Advisor checks, Simulink Test Harnesses, and guaranteed WCET on target platform); automatic tests by continuous integration at push time & issue tracking with GitLab; requirements tracking in Polarion.

Extended F-16 Modelling and Control | [TUM-FSD](#)

📅 05.2015 – 07.2017

DEV Research Project

- **Role:** Lead of simulation model and baseline control design
- **Main Contribution:** Derive & build the simulation model and closed-loop framework. Verified the INDI concept and compared several different control allocation schemes.
- **Objective:** The highly redundant and nonlinear model serves as a conceptualization platform before a strategy is utilized to industry projects.
- **Development:** The closed-loop framework is setup in MATLAB/Simulink with Flight-Gear visualization. Implementation details are found in Publications [1][3].

Dragonfish - 5kg Tilt-rotor Transition UAV | [Autel Robotics Europe GmbH](#)

📅 06.2016 – 01.2020

DEV Industry Project

- **Role:** Lead of control allocation development
- **Main Contribution:** Developed the control allocation scheme for the entire envelope. Deeply involved in the autopilot design of attitude and velocity control, debugging and tuning the closed-loop behavior. The controller provides a smooth transition and unified behavior between hover and wingborne flight.
- **Development:** Deploys MATLAB/Simulink and code generation tool chain with some hundred hours of on-site flight test and tuning. By spring 2020, the complete autopilot software is matured for product pre-release.
- In addition, extensively contribute to administrative communication processes in the cooperation between the university team and the Chinese headquarter.

Manta Ray - 30kg eVTOL Delivery Drone | [Phoenix-Wings GmbH](#)

📅 06.2020 – Now

DEV Industry Project

- **Role:** Programmer of Linearization Tool in C++
- **Main Contribution:** Design and implement the linearization tool in C++, as well as the MATLAB/Simulink interface for model verification and further design tasks.
- **Development:** Use generated C code of the Simulink model as interface. Code in Linux OS with open source Eclipse IDE and CMake projects. Use Ipopt and lapack library as trim solver.
- Additionally participate in sensor calibration processes and extensive software and on-site flight testing for the product tool chain; support in administrative processes with Chinese customers.

FIRST AUTHOR PUBLICATIONS

1. *Attainable Moment Set Optimization to Support Configuration Design: A Required Moment Set Based Approach*
2. *Saturation Protection with Pseudo Control Hedging: A Control Allocation Perspective*

3. *Control Allocation Framework with SVD-based Protection for a Tilt-rotor VTOL Transition Air Vehicle*
4. *Modeling and Incremental Nonlinear Dynamic Inversion Control for a Highly Redundant Flight System*
5. *Control Allocation Framework for a Tilt-rotor Vertical Take-off and Landing Transition Aircraft*